

Pragati Prithvi

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About

Electronics and Computer Engineering graduate (CGPA 9.5/10) with experience in embedded systems, PCB design, hardware validation, and microcontroller-based development. Through internships, rover development, and technical leadership roles, I have worked on hardware-software integration, communication protocols, and system debugging. When I'm away from a workbench or PCB, you'll usually find me reading, listening to music, or feeding street dogs

Education

SRM Institute of Science and Technology (SRMIST)

B.Tech – Electronics & Computer Engineering | CGPA: 9.5/10

2022 – 2026
Chennai, Tamil Nadu

Academic Merit Scholarship (all semesters) • Best Performer Recognition

Technical Skills

Embedded Systems	STM32, ESP32, ESP8266, Arduino, Embedded C, RTOS Basics, CAN, SPI, I2C, UART
Hardware Design	Schematic Design, PCB Layout, Altium Designer, KiCad, Hardware Bring-up, SMD Soldering
Hardware Debugging	Oscilloscope, Logic Analyzer, Multimeter, Signal Validation, Fault Analysis
Programming	Embedded C, C++, Python, MATLAB
Version Control	GitHub Actions
MBD	MATLAB, Simulink, Simscape, Stateflow

Experience

Model-Based Developer Intern – BMS

Sep 2025 – Jun 2026

Euler Motors, Delhi, India

- Built and extended Simulink models for Battery Management System (BMS) subsystems to support hardware-software co-simulation and system-level validation in an automotive context.
- Automated end-to-end model validation workflow using Github Actions, reducing manual test overhead by enabling automatic regression runs on every model push; developers upload models, testers define test cases, pipeline orchestrates execution and reports results.
- Developed and validated Battery Management System (BMS) protection and monitoring logic in Simulink, collaborating with firmware and systems teams to ensure functional correctness and hardware correlation.
- Automated simulation runs, parameter sweeps, and regression tests; generated dashboards and concise summary reports using MATLAB and Python to communicate results to cross-functional teams.
- Investigated hardware-software integration issues, performed root-cause analysis, and validated system behaviour against real hardware test scenarios.

BMS Hardware and Testing Intern

Jun 2025 – Jul 2025

Gajari Brisk (EV Startup), MNNIT, Prayagraj

- Conducted hardware testing and verification of Li-ion BMS circuits; measured latency and utilization of protection and cell-balancing circuits using oscilloscopes and lab test equipment.
- Prepared and executed validation test plans for SoC/SoH estimation and protection circuits; documented assumptions, confidence levels, and test reports aligned with design validation workflows.
- Performed circuit-level fault analysis and root-cause investigations; contributed to improving fault-handling coverage across the BMS hardware architecture.

Projects

Rudra Rover – R25

Altium Designer, STM32, CAN/I2C/UART, Embedded C, MATLAB

- Designed and developed custom multi-layer PCBs for rover electronics including power distribution, sensor interfaces, motor-control subsystems, and communication networks.
- Performed board bring-up, debugging, and validation of STM32 and ESP32 based embedded systems using oscilloscopes, logic analyzers, and serial debugging tools.
- Integrated sensors, actuators, and distributed controller nodes using CAN, I2C, and UART communication protocols; verified system reliability during field testing and competition environments.
- Developed embedded firmware in C for peripheral interfacing, sensor acquisition, communication handling, and subsystem control.

ArchEV – Autonomous Robotic Charging Arm

Linux, ROS2, Embedded Systems, Computer Vision, PCB Design, ESP32

- Integrated camera-based charging-port detection and validated image acquisition pipelines on a Linux-based robotics platform.
- Performed hardware bring-up, debugging, and system-level validation across mechanical, electrical, and software subsystems.

AgroSense – Smart Crop Monitor

Dual ESP32, Custom PCB (EasyEDA), Python, ML

- Designed a custom dual-ESP32 PCB integrating soil moisture, temperature, humidity, and NPK sensors with real-time data acquisition and wireless communication.
- Implemented sensor interfacing, communication handling, and hardware validation across multiple environmental test conditions.

Extracurricular Activities

Technical Director – Electronics

Sep 2023 – Apr 2026

Rudra SRM Mars Rover Team, Chennai

- Led a 6-member electronics sub-team responsible for embedded hardware design, PCB development, board bring-up, subsystem integration, and field validation of rover electronics.
- Integrated STM32 and ESP32 based control nodes over CAN, I2C, and UART networks; supervised debugging, testing, and deployment of mission-critical embedded subsystems.

Club Head

Apr 2023 – Apr 2026

SRM Electronics Club, Chennai

- Led 5+ embedded systems projects (Arduino, ESP32); organised PCB design, MATLAB/Simulink, and Embedded C workshops reaching 100+ students.
- Coordinated 50-member team across IoT innovation challenges and inter-college technical competitions.